



Permeability Dependence on Particle Size of Alloy Based FeSiCr Metal Powder



Y. J. Choi, M. Y. Lee, H. J. Woo, K. W. Cho, B. W. Lee*

Department of Physics and Oxides Research Center, Hankuk University of Foreign Studies, Yongin 17035, South Korea

Introduction

- ❖ FeSiCr powder is a widely known material for high frequency power inductor material.
- ❖ Minimization, high currentization, high efficiency are three trends of power inductor development.
- ❖ In order to minimize the power inductor, we should gain high permeability.
- ❖ For high currentization, good DC-bias characteristics are required, and these are proportional to M_s .
- ❖ To have a high efficiency in the device, the core loss should be low.
- ❖ Core loss consists of hysteresis loss and eddy current loss. In order to reduce the hysteresis loss, we should use a material with low coercivity(H_c), and for low eddy current loss, the electrical resistivity should be high.
- ❖ In our research, we divided the FeSiCr powder in diverse particle sizes and found that the particle size can affect on the permeability of the material.

Experiment

- ❖ FeSiCr powders were divided in six sections of particle sized by sieving method.
- ❖ Crystalline structure was studied by XRD.
- ❖ Magnetic properties were studied by using VSM.
- ❖ 3g of powder was mixed with 0.3wt% of epoxy and pressed in 3 tons for 1 minute.
- ❖ Toroidal core with inner diameter 13mm and outer diameter 20mm was fabricated as the sample for Impedance Analyzer.
- ❖ Permeability, inductance, Q-factor were measured by Impedance Analyzer.

Results & Discussion

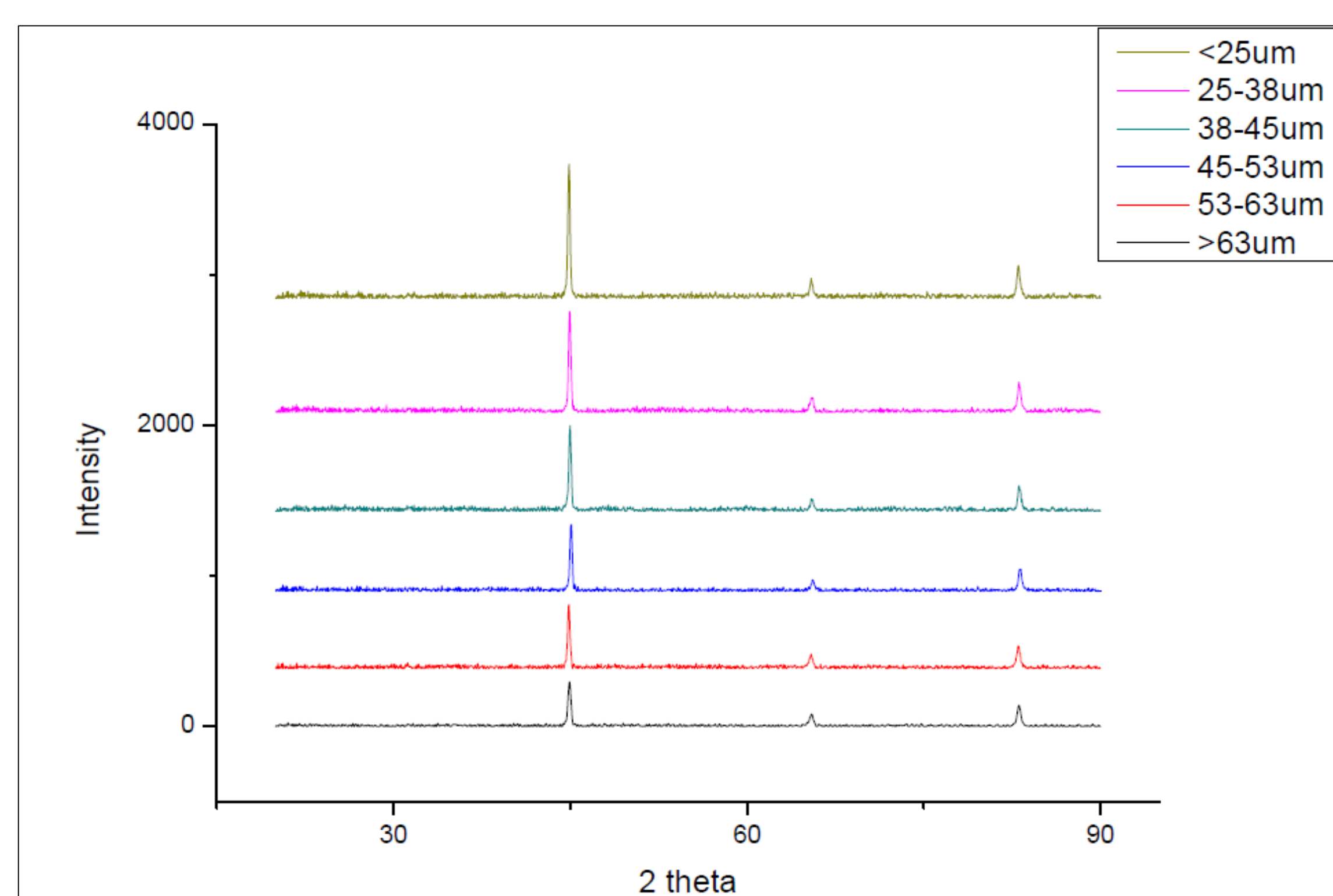


Fig 1. XRD patterns of FeSiCr alloy

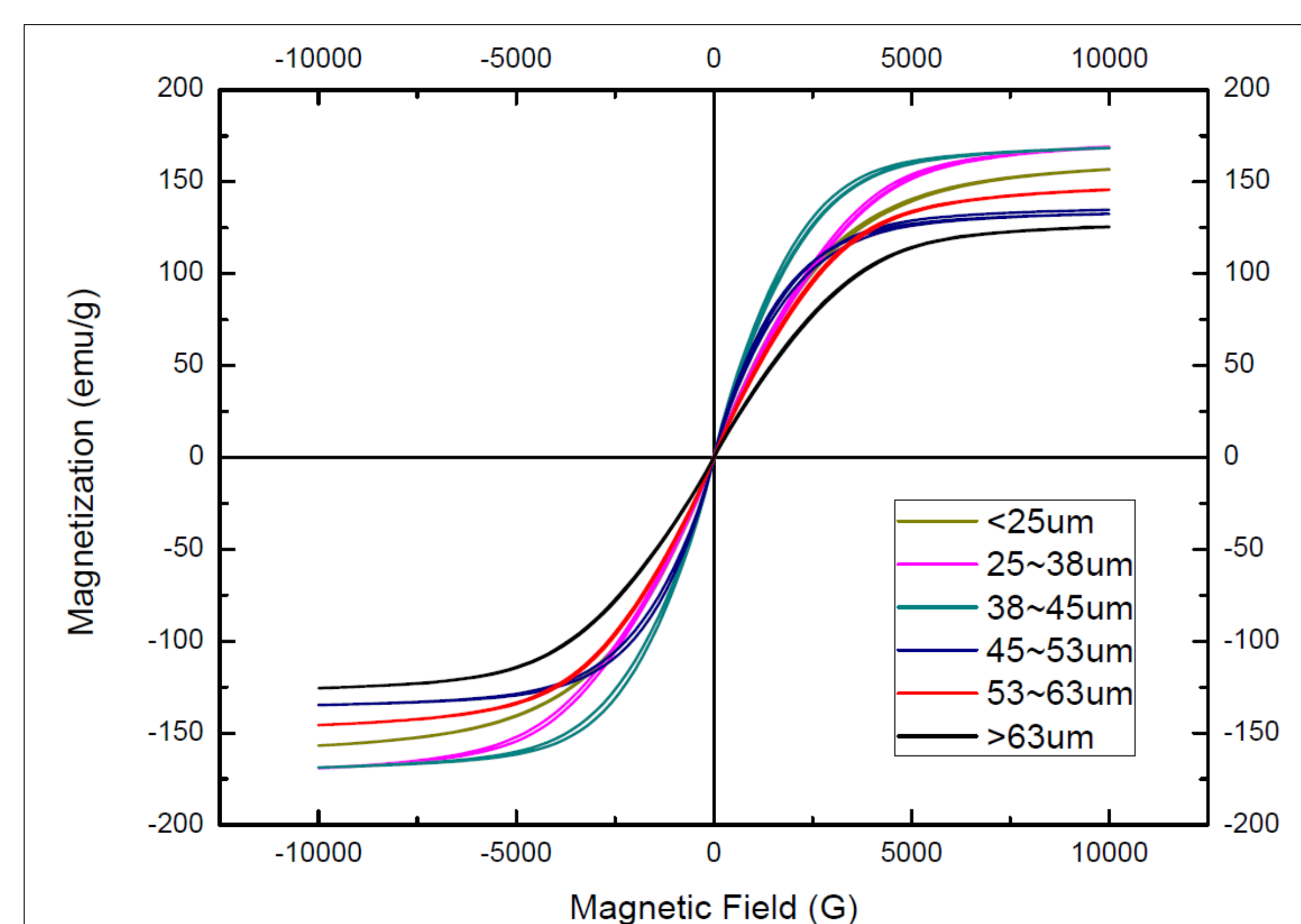


Fig 2. Hysteresis loop of FeSiCr alloy

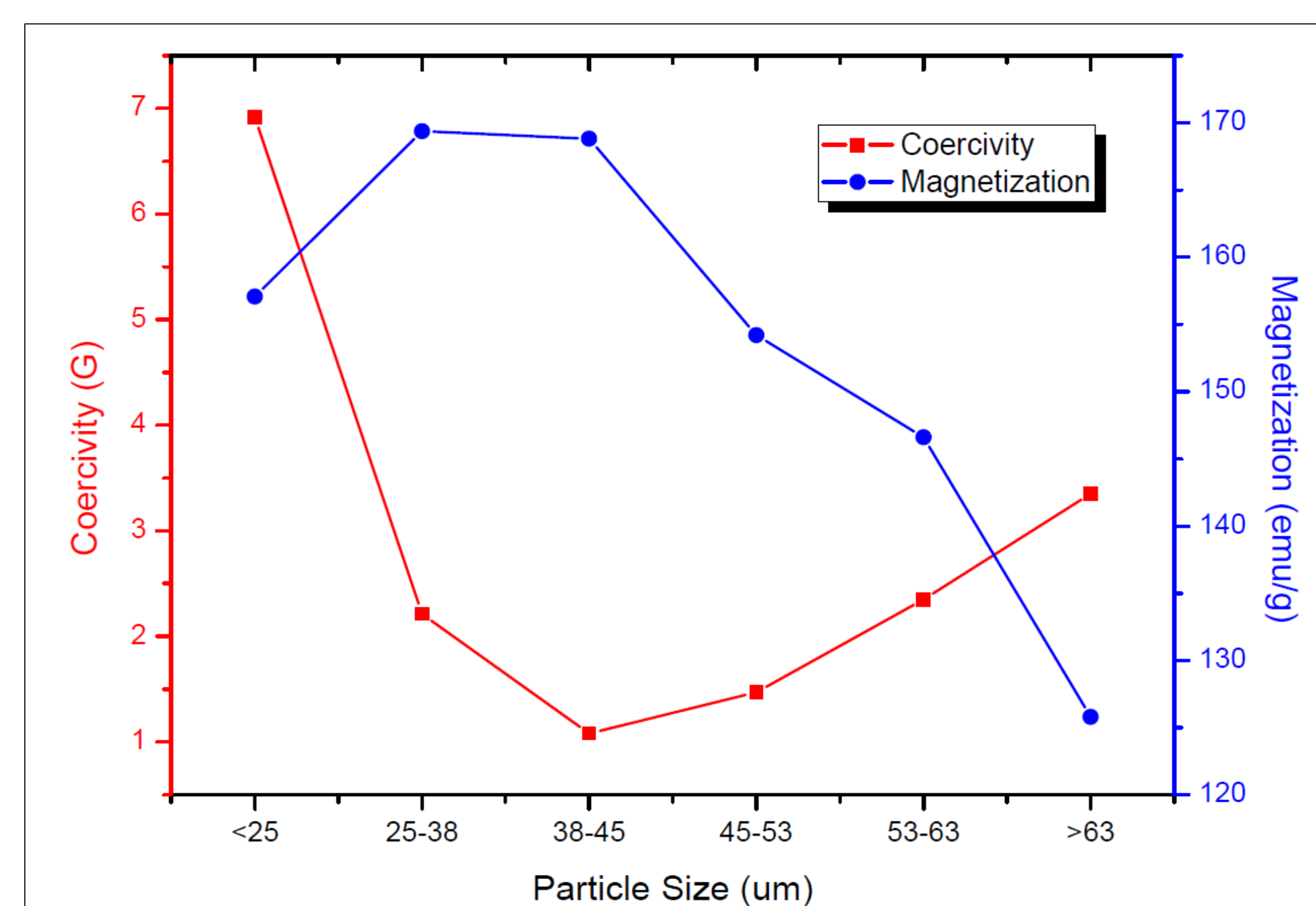


Fig 3. H_c and M_s of FeSiCr alloy

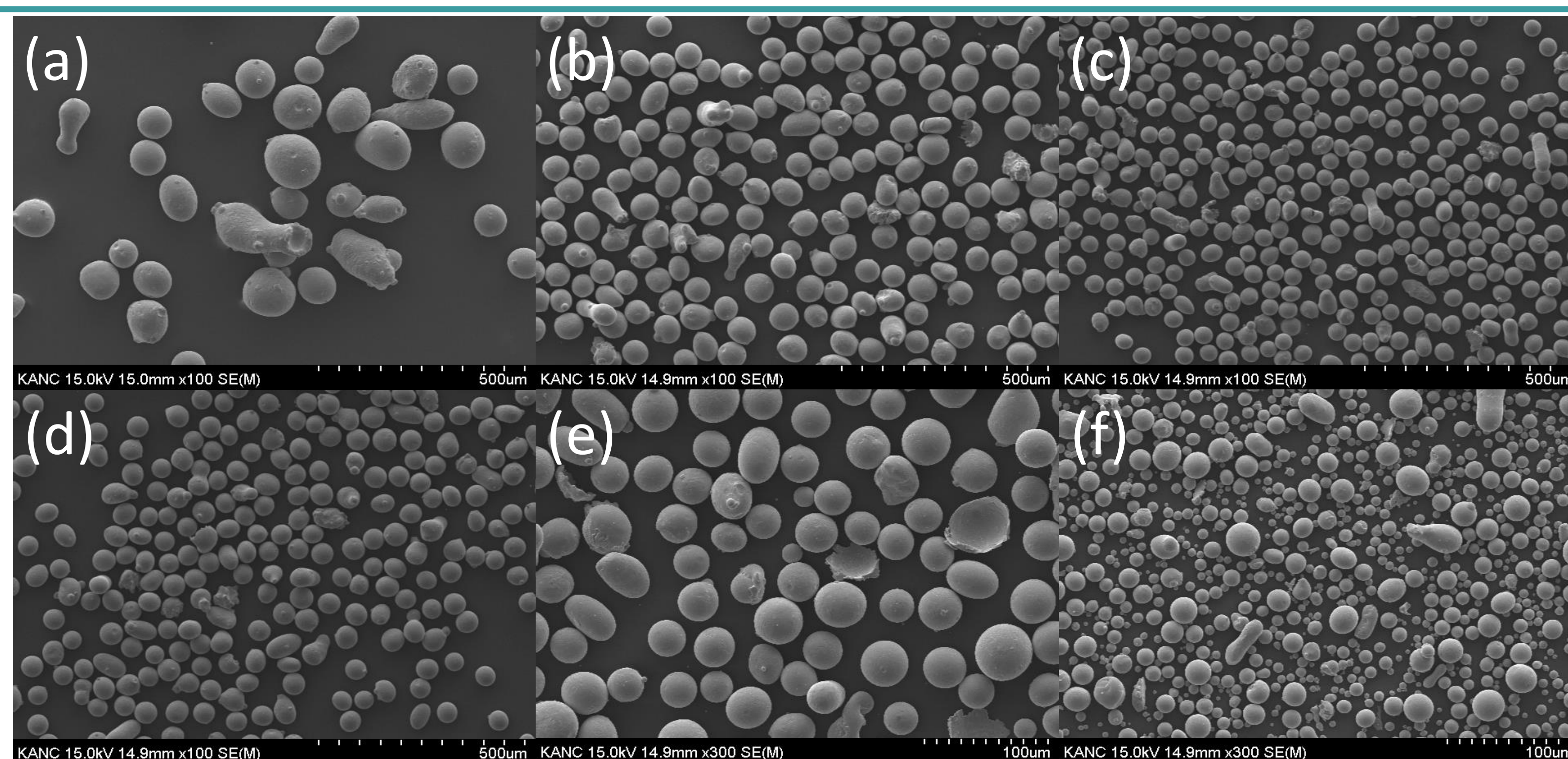


Fig 5. SEM images of FeSiCr alloy (a) $>63\mu\text{m}$, (b) $53\sim63\mu\text{m}$, (c) $45\sim53\mu\text{m}$, (d) $38\sim45\mu\text{m}$, (e) $25\sim38\mu\text{m}$, (f) $<25\mu\text{m}$

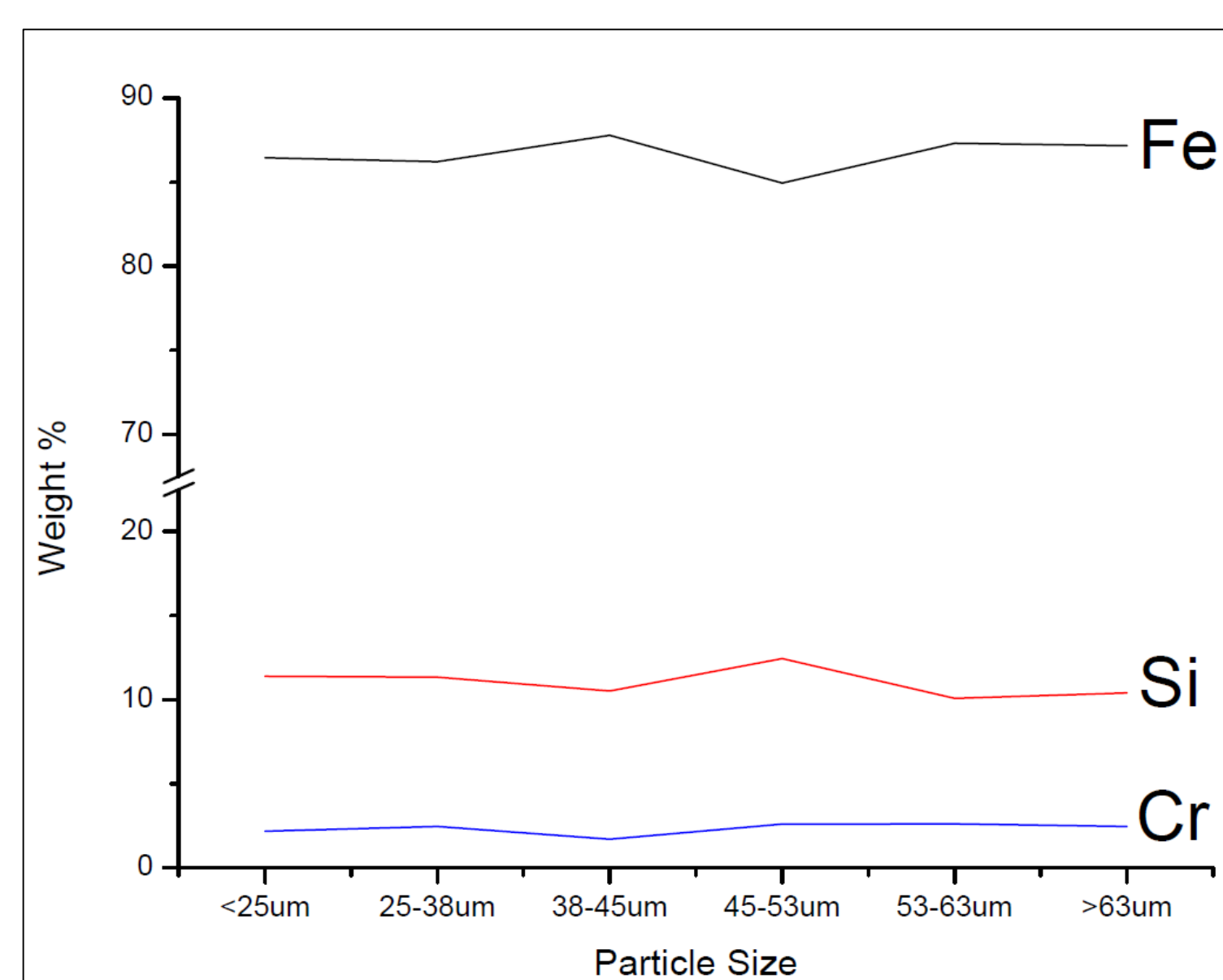


Fig 6. EDS data of FeSiCr alloy

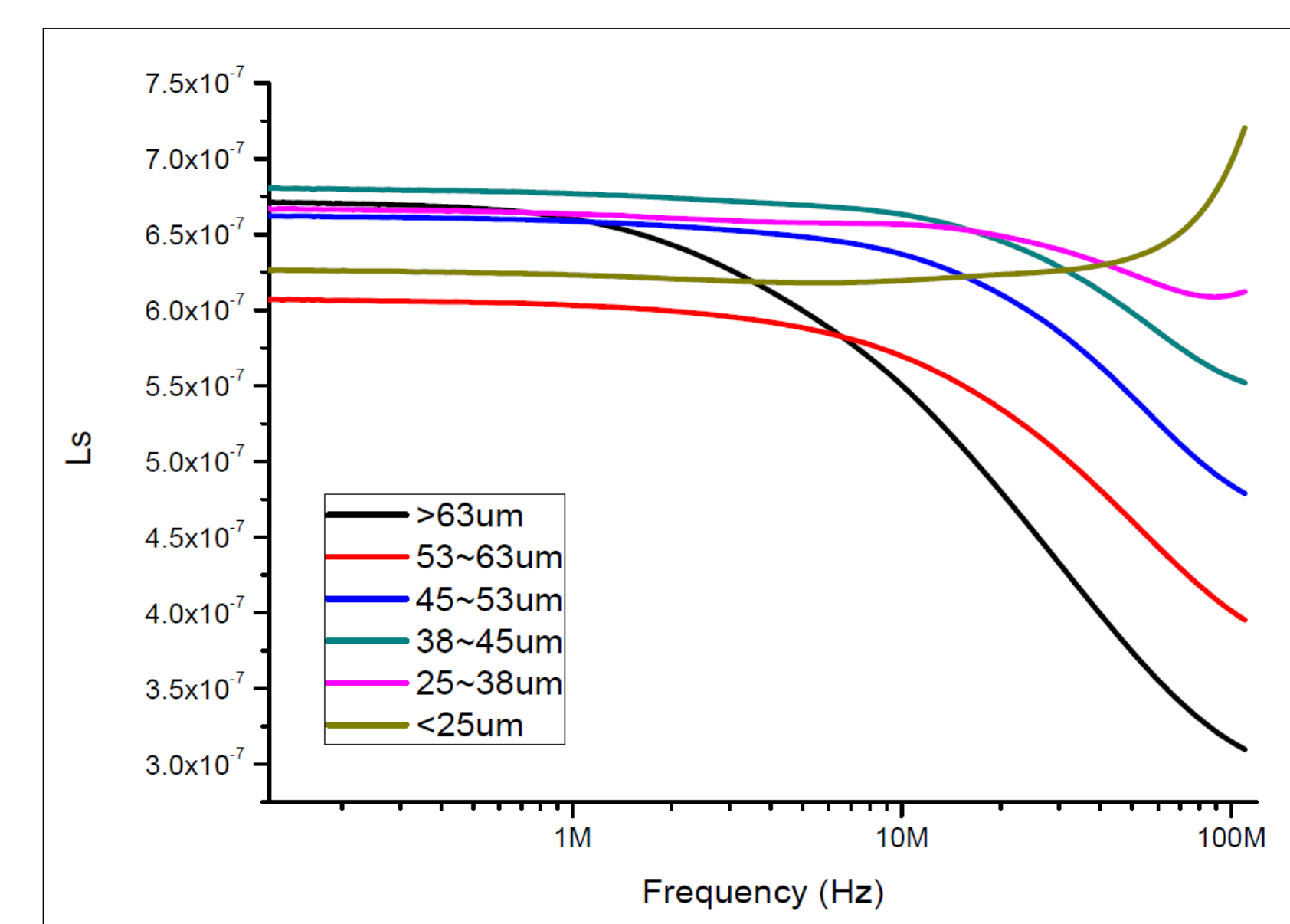


Fig 7. Impedance of various particle sizes of FeSiCr alloy

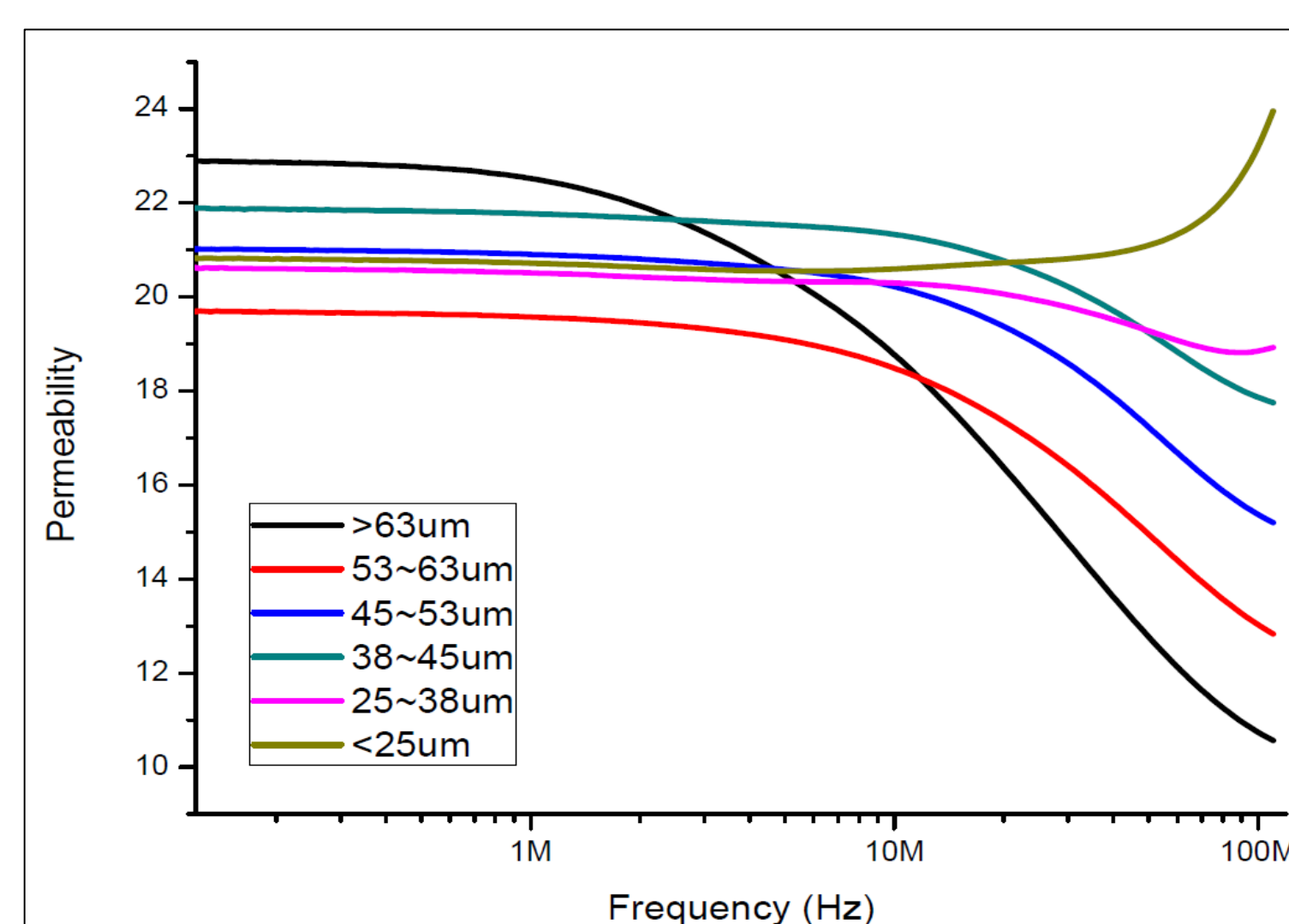


Fig 8. Permeability of various particle sizes of FeSiCr alloy

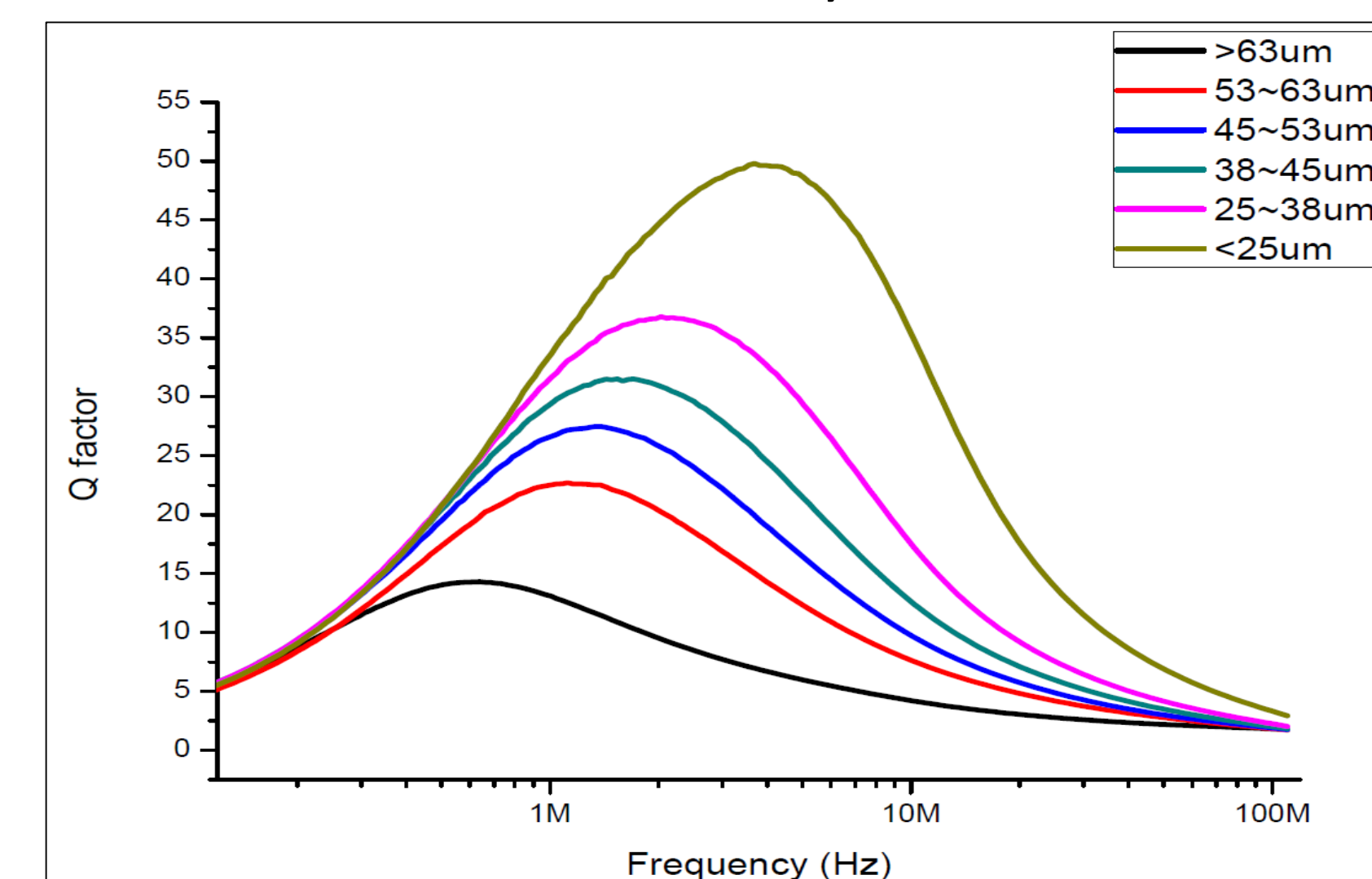


Fig 9. Q-factor of FeSiCr alloy

Summary

- ✓ XRD patterns shows that the peaks of FeSiCr alloy are same as Fe.
- ✓ VSM data shows that FeSiCr alloy has a soft magnetic property and the size with $38\sim45\mu\text{m}$ has the best soft magnetic characteristic.
- ✓ The SEM image shows that the powder has a cylindrical shape, and this means that the powder was made by gas atomization.
- ✓ The highest permeability is shown at the size of $>63\mu\text{m}$, but it rapidly decreases as frequency increases. So, $38\sim45\mu\text{m}$ size of powder seems better for power inductor material
- ✓ $<25\mu\text{m}$ of particle shows the highest Q-factor and resonance frequency.